CYBER SECURITY · INTERNSHIP PROJECT

Personal Firewall Security Dashboard

A Python-based graphical application for real-time network monitoring, firewall rule management, security alerting, and activity logging — developed as part of a cybersecurity internship by **Shivansh**.

Introduction

A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. Firewalls play an important role in protecting systems from unauthorized access, malicious activities, and network-based attacks.

This project implements a Personal Firewall Security Dashboard using Python. The application provides a graphical interface that allows users to monitor active network connections, manage firewall rules, analyze traffic, and receive security alerts.

The project was developed to understand the fundamentals of network security, traffic monitoring, logging mechanisms, and firewall rule management.

Project Objectives

The main objectives of this project are:

1. **Develop a Personal Firewall**

   To develop a Personal Firewall application using Python.

2. **Monitor Real-Time Connections**

   To monitor real-time network connections.

3. **Manage Blocked IPs & Ports**

   To manage blocked IP addresses and ports.

4. **Generate Security Alerts**

   To generate security alerts based on network activity.

5. **Maintain Activity Logs**

   To maintain logs for monitoring and auditing purposes.

6. **Graphical Dashboard**

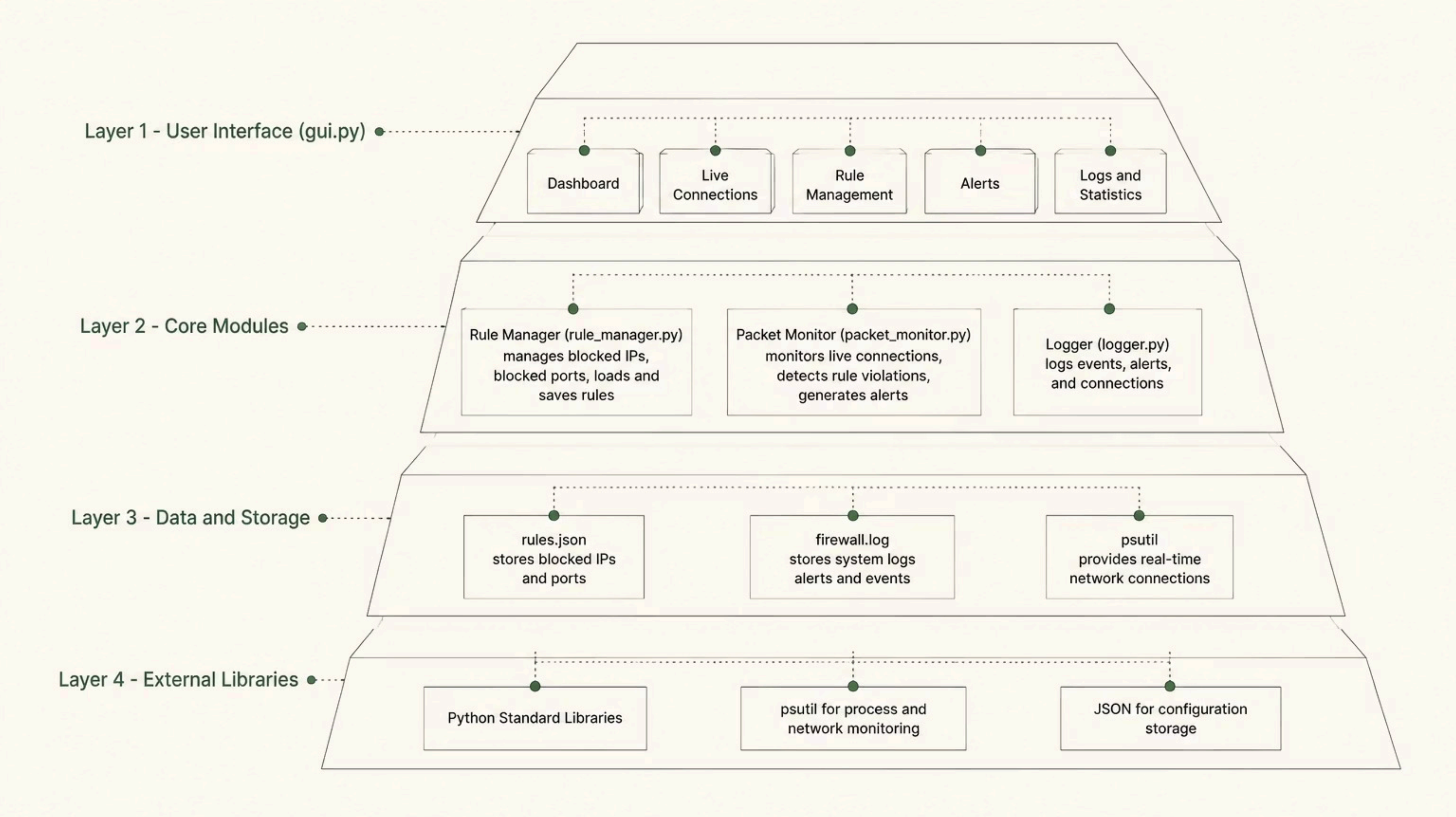
   To provide a user-friendly graphical dashboard for firewall management.

Technologies Used

Technology	Purpose
Python	Core Programming Language
Tkinter	Graphical User Interface
Psutil	Network Monitoring
JSON	Configuration Storage
Git	Version Control
GitHub	Project Hosting

System Architecture

The system consists of four major components organized in a layered architecture, from the user interface down to data storage and external libraries.



GUI Module — gui.py

Provides the graphical interface for:

- Viewing live network connections
- Managing firewall rules
- Viewing security alerts
- Monitoring statistics

Rule Manager — rule_manager.py

- Adding and removing blocked IP addresses
- Adding and removing blocked ports
- Loading and saving rules

Packet Monitor — packet_monitor.py

- Monitoring active network connections
- Detecting blocked ports and IP addresses
- Generating alerts

Logger — logger.py

- Logging events and recording alerts
- Maintaining activity history

Project Structure

The project follows a clean, modular directory structure separating source code, configuration, logs, and tests for maintainability and clarity.

```
Personal-Firewall/
├── src/
│   ├── firewall.py
│   ├── gui.py
│   ├── logger.py
│   ├── packet_monitor.py
│   ├── rule_manager.py
│   └── __init__.py
├── config/
│   └── rules.json
├── logs/
│   └── firewall.log
├── tests/
│   ├── test_logger.py
│   ├── test_monitor.py
│   └── test_rules.py
├── screenshots/
├── requirements.txt
├── README.md
├── LICENSE
└── .gitignore
```

Features Implemented

Real-Time Network Monitoring

The application continuously monitors active network connections using the Psutil library.

Firewall Rule Management

Users can add and remove blocked IP addresses and blocked ports through the dashboard.

Security Alerts

The dashboard generates alerts whenever a blocked IP address, blocked port, or suspicious activity is detected.

Activity Logging

All important events are recorded in log files for future analysis.

Dashboard Statistics

The dashboard displays active connections, firewall rules, security alerts, and system uptime.

Implementation Details

Rule Storage

Firewall rules are stored inside a JSON configuration file.

```json
{
  "blocked_ips": [],
  "blocked_ports": [21, 23, 445]
}
```

Connection Monitoring

The Packet Monitor uses the **Psutil** library to retrieve active network connections in real time.

Logging

The Logger module records:

- Information messages
- Warning messages
- Error messages

All logs are stored inside `logs/firewall.log`.

Screenshots

Key views from the Personal Firewall Security Dashboard interface.

The dashboard displays active connections, blocked IPs, blocked ports, and security alerts in a unified interface. The architecture diagram above shows the four-layer system design.

Advantages, Limitations & Future Roadmap

Advantages

- Graphical user interface for firewall management
- Real-time monitoring of active network connections
- Rule-based detection of blocked IP addresses and ports
- Persistent rule storage using JSON configuration
- Event logging and alert generation
- Modular, extensible codebase

Limitations

- Does not perform packet-level inspection or deep packet analysis
- Limited to monitoring active connections; does not filter packets at kernel level
- Rule management is manual; no automatic threat detection or machine learning
- No integration with external threat intelligence feeds
- No cloud-based or remote monitoring capabilities
- Designed as an educational project, not for production use
- Requires administrator or elevated privileges for comprehensive network monitoring

Future Enhancements

Packet-Level Inspection

Use libraries such as Scapy for deeper traffic analysis.

Enhanced Logging

Adopt structured formats such as CSV or SQLite for improved traceability.

Automated Rule Suggestions

Derive recommended rules from observed traffic patterns.

Kernel-Level Filtering

Integrate system-level firewall APIs for packet filtering support.

Alert Notifications

Enable email or webhook-based notifications for critical events.

Web-Based Dashboard

Provide local network access through a browser-based interface.

Test Coverage Expansion

Increase unit and integration tests for reliability and maintainability.